Chem 4410 Syllabus

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Course Description

This course examines the qualitative and quantitative **relationships** between the **rates and mechanisms** of organic reactions, and the **electronic and physical structures** of reactants. Among the topics considered are: theory and applications of inductive and resonance effects, **linear free energy relationships**, kinetic isotope effects, solvent effects, steric effects in substitution and elimination reactions, acids and bases and pericyclic reactions, applications of semi-empirical and *ab initio* molecular orbital calculations.

Prerequisite: Chemistry 2420, 3420

Where & When

We are scheduled to meet at 8:30–9:20 in KCI 202 MWF.

Instructor

Dr. Barry Linkletter

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PhD. – McGill University (Jik Chin)
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Office Hours

I have an "open door policy" so please come by whenever I am available. However, if you want to be sure to meet at a particular time just contact me to make an appointment.

Textbook

The textbook for this course is an **excellent book** and worth getting a copy and keeping it forever. There is a chapter for almost every field of chemistry in it. I will be using

> **Modern Physical Organic Chemistry** by Eric V. Anslyn and Dennis A. Dougherty. University Science Books.
> MIT Press Webpage
>
> **Note**: The publisher of this textbook, *University Science Books*, has recently been **purchased** by the *MIT Press*. The website has **changed** and some links to the textbook in this webbook may be broken. Please **let me know** if you find a broken link.

Access to the textbook material can be found in the links below or in the actual physical textbook.

- **UPEI Bookstore**:
  Order textbook from the UPEI Bookstore
- If that link does not work, you can purchase the **e-book** directly from VitalSource.
- A copy is available on reserve at the library.
- A copy is available on the shelves in the **chemistry lounge**. Please leave it there **for all to use**.
- **Front Matter of Textbook** (Moodle). This is the table of contents and the **preface** from the textbook. It was available on the former publisher's website. MIT press no longer provides it and I have made a copy available in the course webbook,
- **Appendix 5 Problems** (Moodle). I have copied these **questions from the textbook** and made them available in the course webbook. Please obtain access to the textbook soon as I won't be able to do this again. Respect copyright—someday you may be an author.

Other Textbook Suggestions

The following books are also helpful:

- **McMurry's Organic Chemistry**
  OpenStax Organic Chemistry
  A free textbook for 2^{nd}-year organic chemistry.
- **Physical Organic Chemistry**, by Neil Isaacs. Available in the UPEI Library stacks: **QD476.I846**

- **Organic Chemistry: An Intermediate Text**
  Wiley Online Library via UPEI library online holdings
  by Robert Hoffman. A good source for reviewing 3rd-year advanced organic chemistry.
- **Mechanism and Theory in Organic Chemistry**, 3rd Edition, by Lowry and Richardson. Available in the UPEI Library stacks: **QD476.L68**
- **Structure and Mechanism in Organic Chemistry**, by Felix A. Caroll
- **Advanced Organic Chemistry: Reactions and Mechanisms**, by Bernard Miller; especially Chapters 1 and 5.
- Oxford Chemistry Primers #81: **Structure and Reactivity in Organic Chemistry**, by Howard Maskill.
- Oxford Chemistry Primers #45: **Mechanisms of Organic Reactions**, by Howard Maskill.

Method of Delivery

The course will be presented in a **topic format**. Most lesson topics will span three to six classes except where holidays or storms interfere. The first subject is review and will span the first five class meetings.

Course materials and communications are available through Moodle. The course will be organized in a topic format with three or more **in-person** class meetings.

Evaluation

Grading will follow the scheme outlined below. This grading scheme is open for revision. Any ideas?

Grading Scheme

Component	Grade
Class Participation	10%
Assignments	40%
Personal Exploration Report	20%
Mid-Term Tests	30%

Active **class participation** is required in this course. Students are expected to arrive prepared to participate in discussion of the topic. **Readings and exercises** will be assigned in advance.

The **assigned** textbook **questions** are meant to stimulate discussion and develop your **skills**. You may work together on problems, just thank those who helped you when presenting your solution. Take note that **the assignments are to be completed individually**.

Each week will feature an **assignment** or other goal to be completed. Most of these assignments/goals will involve writing a short **report**. Students will have at least 4 days to **complete** any assignment and **submit** the report.

Late reports may not be accepted unless **arrangements** are made in advance with a valid reason.

During the course, you will begin an exercise called the **personal exploration**. You will choose a **topic** that features physical organic chemistry and **explore** the current **literature**. This exercise will be **developed** over four weeks of the course and, in the end, you will produce a major **report** on your activities.

Mid-Term Test Schedule

Event	Date
Test #1	Oct. 26^{th}
Test #2	Nov. 23^{th}

There are **two mid-term tests** planned. They will be written during class time. There is no **final exam**.

I am open to adjustments of the grading scheme. Suggestions are welcome.

Academic Integrity

Academic integrity is important. As a **community of scholars**, the University of Prince Edward Island is committed to the principle of academic integrity among all its participants.

Academic **dishonesty** as defined in Academic Regulation 20 will not be tolerated.

The Department of Chemistry follows the principles of **progressive** discipline in response to concerns surrounding **academic integrity**.

For **reports and assignments**, the first incidence of academic misconduct will result in a verbal warning with a report to the Department Head. Students will have 3 days to resubmit the assignment for a grade up to 50% of the original mark.

A **second incidence** of academic dishonesty will result in a grade of zero on the assignment with a report to the Dean of Science. A **third violation** will result in zero in the course.

All students must have a current Academic Integrity Moodle Badge to receive a grade. Nothing will be graded unless the **badge** is complete.

For **tests or exams**, the **first** incident of academic dishonesty will result in zero for the assessment and the **second** will result in zero for the course. Both will result in a report to the Dean of Science.

Appeals

If you feel that you have been treated unfairly in a course there is an avenue for appeal. The Department of Chemistry has an **academic appeals policy** that is available from the departmental office.

Accommodation Statement

Students may request **accommodation** as a result of **barriers** experienced related to disability, or any characteristic protected under Canadian Human Rights legislation.

Students who require academic accommodation for either classroom participation or the writing of tests and exams should make their request to **Accessibility Services** as soon as possible.

Please visit UPEI Accessibility Services for more information.

Land Acknowledgement

The land on which the University of Prince Edward Island is located is **Epekwitk**, the traditional and current territory of the **Mi'kmaq People** of this region. It is one of seven traditional Mi'kmaq regions that comprise Atlantic Canada, parts of eastern Quebec, and Maine.

Mi'kma'ki is the ancestral and unceded territory of the Mi'kmaq people, who, in 1725 first signed the **Treaties of Peace and Friendship** with the British Crown. Those treaties did not deal with the surrender of lands and resources, but instead recognized Mi'kmaq title and negotiated a path toward an ongoing relationship between nations.

We acknowledge that we carry out our daily work and learning in Mi'kma'ki.

Storm Cancellations

If UPEI is closed for inclement weather, no in-person class will be held that day. Should a storm cancellation fall on the date of an examination or assessment, the date will be moved to the next available class.

Exam Absences

If you miss a test or exam because of illness, death in the family, etc., please let the instructor know **as soon as possible** so that alternate arrangements can be made.

Communications

UPEI business is conducted by email. Your **UPEI email address is your official point of contact**. Please check your UPEI email often.

Emails will be responded to during standard working hours, 9 am–5 pm.

Please provide instructors, coordinators, teaching assistants, and staff with at minimum 48 h to **respond** before sending a follow-up email.

Outcomes

One must approach all endeavours with a **set of goals**. Here are the expected outcomes of a university education, the chemistry program, and this course.

University Outcomes

Students who have completed a **bachelor of science degree** will have achieved the following goals:

- Be able to **gather information** through a well documented process of **inquiry**.
- Be able to **analyze and interpret** that information and produce a **conclusion.**
- Be able to effectively **communicate** the conclusion, the original data and the methods used in the analysis.
- Have mastered **tools and methods** to accomplish the above within their field of endeavour.
- Be able to work with others demonstrating **leadership and collaborative skills** and to function as a member of an interdisciplinary **problem solving** team.
- Have become **independent thinkers** who are responsible for their own learning.
- Have demonstrated the principles of **academic integrity** in all their scholarly activities.

Program Outcomes

Students who complete a **BSc in Chemistry** will have achieved the following goals:

- Have developed an understanding of the principles of various **fields of chemistry** (organic, inorganic, physical, analytical, and biochemistry).
- Be able to **safely conduct oneself in a chemical laboratory** and any location where chemistry is applied.
- Be able to **design and carry out scientific experiments**, accurately **document** methods and **analyze** the results of such experiments.

- Be able to **use a variety of modern instruments** for separating and analyzing mixtures and for characterizing molecules and materials and to **apply these skills** to appropriate chemical problems.
- Be able to **interpret** the modern primary **literature** of chemistry and to **communicate** the results, conclusions, and relevance of scientific experiments to a specific audience in a variety of formats, including **peer-review** publications.
- Understand the **role of chemistry** in making the world around us better for everyone and understand the risks and harms that can be caused by chemistry.
- Be able to **explore new areas of research** in both chemistry and allied fields of science and technology.
- **Honours students** will have greater exposure to upper-level chemistry courses and research under a supervisor with the goal of producing a research **thesis** and publicly presenting the results.

Course Outcomes

Students who successfully complete this course of study in **physical organic chemistry** will have achieved the following goals:

- Be able to **describe molecular structure** by the use of valence bond theory and molecular orbital theory.
- Be able to use **computational tools** to determine optimized structures of small molecules and interpret the energetics of simple reactions.
- Be able to **design experiments** to determine the kinetic and thermodynamic properties of a reaction and to **interpret the results**.
- Be familiar with different types of **reaction mechanisms** and be able to describe different types of **reactive intermediates**.
- Be able to **propose a reaction mechanism** after examining the reactants and products of a reaction.
- Be able to use reaction kinetics as a tool for **understanding reaction mechanisms.**
- Develop a deeper understanding of **acid/base equilibria** in aqueous and non-aqueous systems and determine whether a reaction is **acid or base catalyzed**.
- Be able to apply fundamental concepts of **chemical and biochemical catalysis.**

- Be able to **investigate** a hypothetical reaction mechanism by:
  - using the **Hammett equation** as a tool in studies of organic reactions;
  - using other **linear free energy relationships** (LFERs);
  - incorporating **isotopes**;
  - applying a variety of **other methods**.
- Be able to explain **steric and electronic effects** in several important reactions.
- Develop an understanding of the **transition state** and how its structure changes with steric and electronic effects.
- Be better able to choose suitable **solvents and conditions** for a reaction.
- Understand how **parameter systems** developed using physical organic chemistry are used in **quantitative structure–activity relationships (QSAR)** for drug design.
- Have improved **computer skills** in data analysis, chemical investigation, and record keeping.

Self Evaluation

Almost all students in this course will be in the final year of the chemistry program. **Now is the time** to evaluate yourself as you consider the goals for **university and program outcomes**. How do you feel? Do you have a sense that you are well on the way to achieving them? Are there any weaknesses that could be addressed while taking this course?

Also consider the goals for the **course outcomes**. You might feel that you have accomplished some of them already. **Return to this list** every few weeks and check off the goals you feel you have reached or are progressing toward. By monitoring your progress, you will be better able to see where more work is needed to ensure that all goals are met. Never hesitate to **ask for more** in one area or another.

Essential Tools

Before you read any farther, you need to know about the **tools** that are **needed** to succeed in this course. You will need access to the **textbook**, a **computer** with the software listed below, and the **time** to learn on your own and complete assignments.

Your Textbook

This web-book document that is available on Moodle **is not** the textbook, it is a guide to the course with readings, assignments, documents, etc. collected together. The textbook for this course is...

Modern Physical Organic Chemistry by Eric V. Anslyn and Dennis A. Dougherty. MIT University Press.

This book is a tour de force. I encourage you to buy the textbook and **keep it** for life. It covers the basics of physical organic chemistry and then has chapters that **can be applied in just about any field of chemistry**. Supramolecular chemistry? Check. Conductive polymers? Check. Theoretical chemistry? Check.

The textbook is available in the bookstore and other options for textbook access are described above in the course information section. **All chapters and problem assignments will refer to this book**.

The textbook for this course was **written** by two eminent physical organic chemists and then **peer reviewed** and corrected by dozens more. Look around you. How many students are in this class? It's not 200. I guarantee that **no one is making money** from this book. If we do not support efforts like this, we may never see a second edition.

These authors are far too old for a *Patreon* account. Please **do not pirate** textbooks. I know that downloading is easy to do, but someday you may be writing a textbook of your own. Start building your **karma** now.

The Literature

We will be using literature examples and you will be expected to **become familiar with searching and using chemical literature**. I will try to use **only papers that are accessible through the UPEI Library** and its electronic subscriptions. If I use a paper that is only available through interlibrary loan, I will make it available through Moodle.

Papers that you reference in reports or assignments **must be accessible through the UPEI Library system** or by interlibrary loan. If obtained through interlibrary loan, a copy of the paper must be submitted with your work.

We will not pirate papers, regardless of our opinions about scientific publishing and profit-driven publication models.

The **library** provides a useful search tool that can be used to locate references. If the library does not hold a **license** for a source, you will usually be able to obtain it through **interlibrary loan**, provided that you **give yourself enough time**. Begin research projects early.

Here are some useful literature resources:

- ACS SciFinder via UPEI
- Google Scholar
- UPEI Library Chemistry Resource Page

Your Computer

Welcome to the modern world. For the past 20 years, students have **needed computers to succeed** in university.

You will need **access** to a computer with the **tools** listed below. These tools will be **available** on many **campus** computers. I have selected only **free** software options that you can install on your own computer.

We will not be programming. We will be learning how to use software tools. Any coding that appears in the course will be provided in advance, and I likely borrowed it from somewhere on the internet. I am not a coder, I am a **user**, and users are the true heroes.

Below is a list of software that will be available on many campus computers or can be installed on your own computer.

1. **A chemical drawing program**
    - There are many options. I recommend the *Marvin* package from ChemAxon: ChemAxon Marvin.
    - It runs on multiple operating systems.
    - A convenient web-based version is available through *Chemicalize*: Chemicalize.
    - The website provides access to many of the features of Marvin without installing software.
2. **The usual productivity software**
    - You will need a **word processor**, **spreadsheet**, and **presentation program**.
    - A free option is LibreOffice.
    - Office 365 is also available to all UPEI students.
    - In this case, I recommend the commercial option because it is already available through the university.
3. **Python with the Jupyter Notebook environment**
    - We will use **Python** for data analysis and plotting.
    - It is a powerful scientific tool that you will likely continue using throughout your career.
    - We will access Python through the **Google Colab** website, so there is no need to install software locally.

Your Time

This course will require your time. Only you can explore the concepts presented in the lessons by **investing your time**.

You will need:

- **time to read** the textbook and consider the suggested problems;
- **time to struggle** with software tools as we explore data analysis and computational chemistry;
- **time to write** reports and prepare presentations; and
- **time to relax**.

Course Outline

Below is a list of topics and the corresponding textbook chapters.

Week	Lesson #	Topic	Reading
1 & 2	1	Review and Skill Building	Ch. 1, 2 & Appendix 5
3 & 4	2	Acid Equilibrium, Kinetics & Catalysis	Ch. 5, 7 & 9
5 & 7	3	Linear Free-Energy Relationships	Ch. 8
6		Break Week	
8	4	Test #1 & Isotope Effects	Ch. 8
9 & 10	5	Substitution and Rearrangement Mechanisms	Ch. 11 & literature
		All topics below can be changed. What do you want?	
11 & 12	6	Addition and Elimination Mechanisms & Test #2	Ch. 10 & literature
13	7	Pericyclic Reactions	Ch. 15 & literature
14	8	Review and Wrap-up	literature

To achieve our goals we must examine the following ideas.

The Past is Prologue

We will begin with some of the **core ideas of chemistry**, focusing on structure and energy. This will not be merely a review; it will be a **deep dive**. There are things we did not tell you. You are ready at last.

We must first start with an understanding of molecular structure. We will consider the **nature of the chemical bond**. It is not a line on a page, it is an arrangement of electrons and nuclei that form a stable system. Changes in structure can affect the energy and therefore the reactivity. **Structure affects reactivity**.

To fully understand how structure influences reactivity, we must understand the path from reactant to product. We will develop a deeper understanding of the **reaction coordinate** and the **transition state**. We will see how changing the structure of reactants can change the timing of bond changes in the transition state. We will express this via two-dimensional reaction coordinate diagrams and three-dimensional reaction **energy surfaces** as we try to get as close to the truth as simple line drawings will permit.

Seeing Farther

Now we will move into topics that are familiar but include new ideas for interpreting how structure affects reactivity. We will review the mathematics of reaction kinetics and rate laws and apply these to catalyzed reactions.

We will encounter **acid/base chemistry** yet again. The fifth time is the charm. The mathematics of acid equilibria will be explored and we will leave the world of water behind and experience a place where strong acids show their true power. Can you **measure** the *pH* of pure sulphuric acid? Physical organic chemistry will show you how.

Structure and Reactivity

The heart of physical organic chemistry is the **correlation** of one system with another. Changing the structure of substituted benzoic acids changes the pK_a value. Can we measure the **electronic effect** of these structural changes using pK_a values as the score? Can we score the **size** of chemical groups by how they affect the rate of a given reaction? Can we then use these scores to study the **mechanism** of unrelated systems? Yes we can.

Free energy changes in one system may relate to free energy changes in another system. These correlations are called **linear free energy relationships (LFER)** and represent one of the most important contributions of physical organic chemistry.

If we can quantify how structure affects reactivity, perhaps we can quantify how structure affects the activity of drugs. This is the idea of **quantitative structure–activity relationships (QSAR)** and is an important tool in drug development.

Even tiny changes can have measurable effects. Exchanging a hydrogen for **deuterium** may seem insignificant, yet it can change the activation energy of a reaction. Using **isotope effects**, we can investigate reactions through changes that are neither structural nor electronic. This idea is called the **kinetic isotope effect**.

We can **interpret** the details of reaction mechanisms through observation and detective work. Many **clever methods** have been developed, and we will explore several of them.

Molecular Orbitals

To investigate the effect of structural changes on electronic arrangements in molecules, we must consider the quantum nature of chemical systems.

We will use simple linear combinations of **atomic orbitals** to build graphical models of **molecular orbitals** in small molecules. We will then use simple mathematical approaches to estimate **energies and electron densities** in π systems and explore **electron delocalization and aromaticity**.

Explorations & Examples

We can go anywhere. What would you like to investigate?

Each year we will create a body of work that future students can build upon. We will begin with **examples** of physical organic chemistry being used to investigate **interesting mechanistic ideas**. We may explore addition and elimination mechanisms, substitution and rearrangement mechanisms, and pericyclic reactions, **unless you have other ideas**.

We will **go wherever you wish**. Would you like to explore physical organic chemistry as it is applied to a **particular problem**? Would you like to **survey the work** of an important physical organic chemist? We can do anything, and you will document it all in the form of a report.

You will work on your personal exploration of a **topic** that interests you while we **explore** examples of reaction mechanisms described above.

Lecture Plan

Below is a proposed plan for the course this year. **This will change** as I get to know the preferences and interests of the students and as class cancellations occur.

Lesson	Class	Date	Topic	Reading
1	1	Sept. 9	Welcome and Course Introduction, Review of Structure and Sterics	Ch. 1, 2.1, 2.3
	2	Sept. 11	Review of Structure and Molecular Orbitals	Ch. 1.2, 1.3, 2.4
	3	Sept. 14	Review of Mechanisms and Arrow Pushing	Appendix 5
	4	Sept. 16	Review of Transition State Theory and Reaction Coordinates	Ch. 7.1–7.3
	5	Sept. 18	Review of Energy Surfaces	Ch. 7.8, Appendix 5
2	6	Sept. 21	Brønsted Theory and Acidity Functions	Ch. 5.1–5.3
	7	Sept. 23	Structure Effects on Acid Equilibrium	Ch. 5.4
	8	Sept. 25	Reaction Kinetics and Acid/Base Catalysis	Ch. 7.4–7.6, 9.1, 9.2
	9	Oct. 2	Brønsted Plots	Ch. 8.5, 9.3
	10	Sept. 28	pH-Rate Profiles	Ch. 9.2, 9.3
3	11	Oct. 5	LFER: Inductive Substituent Effects	Ch. 8.2, 8.3
	12	Oct. 7	Resonance Substituent Effects	Ch. 8.2, 8.3
	13	Oct. 9	Interpreting Hammett Plots	Ch. 8.2, 8.3
			Break Week	
	14	Oct. 19	Steric Substituent Effects	Ch. 8.4

	15	Oct. 21	Solvent Effects	Ch. 8.4–8.6
	16	Oct. 23	Investigating Reaction Mechanisms	Ch. 8.7, 8.8
	17	Oct. 26	**Test #1**	Classes 1–17
4	18	Oct. 28	Theory of isotope effects	Ch. 8.1
	19	Oct. 30	Interpreting isotope effects	Ch. 8.1
5	20	Nov. 2	Examples of S_N1 & S_N2 substitution	Ch. 11.1 to 11.5, lit.
	21	Nov. 4	Examples of carbocation rearrangements	Ch. 11.8, lit.
	22	Nov. 6	Examples of nucleophilic rearrangements	Ch. 11.9, 11.10, lit.
	23	Nov. 9	Example exploration: Non-classical cations	Ch. 11.5, 14.5, lit.
		Nov. 11	**No Class** – Remembrance Day	
	24	Nov. 13	Example exploration: Non-classical cations	Ch. 11.5, 14.5, lit.
			All topics below can be changed.	
6	25	Nov. 16	Examples of pH-rate profile for aspirin	Ch. 9.1 to 9.3, 10.17
	26	Nov. 18	Examples of acetal formation/hydrolysis	Ch. 10.2, 10.12, lit.
	27	Nov. 20	Examples of electrophilic addition	Ch. 10.1, 10.3 to 10.5
	28	Nov. 23	**Test #2**	Classes 18 to 29
	29	Nov. 25	Examples of enzyme catalysis	Ch. 9.4, 10.2, 10.12
	30	Nov. 27	Examples of a rearrangement	Ch. 11.8 to 11.10, lit.
7	31	Nov. 30	Examples of cycloaddition reactions	Ch. 15.1 to 15.3, lit.
	32	Dec. 2	Examples of electrocyclic reactions	Ch. 15.4, lit.
	33	Dec. 4	Examples of pericyclic rearrangements	Ch. 15.5, 15.7, lit.
	34	Dec. 7	Examples of carbene reactions	Ch. 15.6, lit.
	35	Dec. 9	Wrapup and reflection	

Assessment Plan

Below is a plan for the **assessments** in this course.

Assignment	Due Before	Due Date	Objective	Grade
0	Meeting 2	Sept. 11	First literature exploration & Learning Plan	4%
1	Meeting 5	Sept. 18	Organic chemistry review	4%
2	Meeting 8	Sept. 25	Dogma of physical organic chemistry	4%
3	Meeting 11	Oct. 5	Acid/base catalysis and Brønsted plots	4%
4	Meeting 13	Oct. 9	Hammett plots	4%
Test #1	Meeting 17	Oct. 26	Covers meetings 1–16	15%
5	Meeting 19	Oct. 30	Transition-state structure	4%
6	Meeting 24	Nov. 13	Data analysis and mechanism interpretation	4%
7	Meeting 26	Nov. 18	Personal exploration proposal	4%
Test #2	Meeting 28	Nov. 23	Covers Meetings 18–27	15%
8	Meeting 30	Nov. 27	Data analysis and mechanism interpretation	4%
9	Meeting 33	Dec. 4	First draft of personal exploration	4%
Report	Meeting 35	Dec. 9	Submit final personal exploration report	20%